

For this project, I focused on counties that were the top 20 in California for farming: Butte, Colusa, Fresno, Glenn, Imperial, Kern, Kings, Madera, Merced, Monterey, San Diego, San Joaquin, San Luis Obispo, Santa Barbara, Stanislaus, Sutter, Tehama, Tulare, and Ventura. The date range I chose was January 1, 2024 - December 31, 2024. I used the following daily columns: county, latitude, longitude, time, temperature_2m_mean, et0_fao_evapotranspiration_sum, soil_moisture_0_to_100cm_mean, soil_temperature_0_to_100cm_mean, r, relative_humidity_2m_mean, vapour_pressure_deficit_max, wind_speed_10m_mean, cloud_cover_mean, and precipitation_sum.

There were no real data quality issues, but having to sort through a large amount of columns was difficult. I also had to change the scope of some of my questions, because they ended up being too descriptive and did not allow me to fetch a result.

With this data, I was able to answer several questions pertaining to farming and weather conditions in California.

1. Which county has the highest average wind speed in the entire year?

Tehama County has the highest average wind speed with a speed of 10.91 km/h.

2. How many counties have moderate cloud cover (30-60%) in the entire year?

17 counties have moderate cloud cover between 30-60%.

3. Which county has the highest total of precipitation in the entire year?

Butte County has the highest total amount of precipitation in the entire year with 1211.7 mm.

4. Which month has the lowest average vapour pressure deficit for every county?

- Butte County: November, 0.2
- Colusa County: December, 0.24
- Fresno County: December, 0.27
- Glenn County: November, 0.17
- Imperial County: January, 0.94
- Kern County: December, 0.35
- Kings County: December, 0.39
- Madera County: December, 0.31
- Merced County: December, 0.35
- Monterey County: January, 0.27
- San Diego County: January, 0.23
- San Joaquin County: December, 0.24
- San Luis Obispo County: February, 0.22
- Santa Barbara County: March, 0.37
- Stanislaus County: December, 0.24
- Sutter County: December, 0.21
- Tehama County: December, 0.19

- Tulare County: November, 0.24
- Ventura County: February, 0.2
- Yolo County: December, 0.17

5. Which 5 counties have the lowest total ET0 in the entire year?

- Monterey County: 1129.8200000000001
- Ventura County: 1130.0799999999992
- San Diego County: 1208.60000000000013
- Santa Barbara County: 1240.9699999999996
- San Luis Obispo County: 1254.27

6. Which counties have ideal temperatures (18-22 Celcius)?

Kern County, Kings County, Tulare County, Fresno County, Glenn County, Colusa County, Madera County, Butte County, Merced County, Stanislaus County, Tehama County, and Sutter County have ideal temperatures between 18-22 degrees Celcius.

7. What counties have a moderate average soil moisture, soil temperature, and humidity in the entire year?

Only San Joaquin County has moderate averages in all three categories for the entire year.